

Measuring Physicians and LLM Relevance Alignment in Medical Question Answering

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ABSTRACT

Large Language Models (LLMs) have demonstrated strong performance on medical question answering (QA) benchmarks, including standardized medical exams. Correct answers alone, however, do not guarantee correct reasoning, as a model can reach the right conclusion while attending to input that a clinician would consider unimportant. Here, we evaluated how physician trainees and LLMs estimate sentence level relevance when answering clinical QA items. We obtained annotations on 933 QA pairs from 53 physician trainees, with a subset independently reviewed by 7 board certified physicians, labelling each sentence of the case description as high relevance, low relevance, or irrelevant. We then compared these trainee annotations with LLM derived relevance labels and measured the effect of restricting the input to the sentences each labeller considered relevant. The average sentence level concordance between LLMs and the physician trainee majority vote ranged from 44.9% to 65.9%, while the concordance between physician trainees and board certified physicians was 72.4%, indicating a modest but consistent gap. Removing sentences that physician trainees judged low relevance or irrelevant increased the accuracy of most LLMs on the four benchmarks, and similar gains were observed when each LLM filtered its own input or the input of another model. These results suggest that current LLMs do not consistently prioritize the same case information as physician trainees and that relevance aware filtering can be a useful probe of model reasoning, though the findings are based on multiple choice exam questions and should be interpreted as an evaluation step rather than direct evidence of readiness for clinical deployment.

1 Main

Large language models (LLMs) have shown strong performance across a range of medical tasks¹, with systems like GPT-4² and MedPaLM³ outperforming human averages on standardized medical examinations^{4–8}. Multiple LLM companies including OpenAI and Anthropic launched the specific HIPAA-compliance LLMs. However, high performance on exam-style datasets may overstate a LLM’s generalizability^{9–11}, as many tasks do not reflect the complexity of real-world use cases¹². In human-facing settings, it is crucial to understand how models filter and prioritize relevant information¹³.

Estimation of contextual relevance is a critical aspect in many applications^{14–16}. Techniques such as semantic entropy¹⁷, influence functions¹⁸, context attribution^{19,20}, and evidence inference²¹ have been employed to assess which elements within a context hold the most importance. Despite these efforts, existing relevance estimations are often imprecise and noisy, with models sometimes producing misleading or overly confident assessments that deviate from human judgment²². Even when estimations appear less noisy, model-generated relevance labels may not concord with those of human experts. This gap is particularly concerning in human-facing domains where alignment with expert judgment is necessary^{23,24}.

Models may depend on input features that a clinician would not treat as informative²⁵. Prior work has shown that restricting inputs to relevant content can reduce distraction, streamline evidence integration, and lower memory requirements^{26–29}. Small, clinically unimportant perturbations of medical questions have also been shown to produce statistically significant shifts in model treatment recommendations³⁰. To date, however, little work has examined whether models and physicians agree on which case information is worth attending to, or whether concentrating models on such content measurably improves downstream answers.

Characterizing whether LLMs attend to the same input sentences that clinicians treat as informative is a useful step toward understanding how these models derive their answers^{31,32}. Such comparisons can complement accuracy based evaluation by flagging cases where a correct answer is accompanied by reliance on sentences that clinicians would not regard as decisive^{33,34}. Alignment between model attributed and clinician identified evidence has also been proposed as one of several factors relevant to clinician trust in diagnostic AI tools^{35,36}. Restricting LLMs to sentences that clinicians judge highly relevant provides an auditing lens to probe whether model predictions depend on content that humans consider peripheral^{37–39}. Such dependencies do not always reduce benchmark accuracy, but they can complicate the interpretation of benchmark performance.

A related concern is that LLMs may be trained on data that overlap with public medical benchmarks. Relevance aware filtering does not by itself distinguish memorization from genuine reasoning, but it offers one additional axis along which such effects can be examined. In our setting the four source datasets span release dates from 2020 to 2025, so the potential for training set contamination varies across subsets and we treat this as a confound rather than a controlled variable.

To study how physician trainees and LLMs interpret the relevance of input patient context, we introduce *MedPAIR* (Medical Physician trainee and AI Relevance), a benchmark that pairs sentence level relevance annotations from physician trainees with LLM derived relevance labels on the same QA items. Medical QA datasets are a primary method of evaluating LLM performance in clinical tasks^{40–43}, yet existing benchmarks primarily score final answers and do not track which parts of the case the model or the human relied on^{44–46}. Recent work on explanation generation and qualitative reasoning analyses focuses on the reasoning produced after the input context has already been interpreted⁴⁷.

We collect sentence-level relevance labels on 933 samples from four QA benchmark datasets - *Massive Multitask Language Understanding* (MMLU)-precision medicine (192 QAs)⁴⁸, *Medbullets* (160 QAs), *JAMA Clinical Challenge dataset* (295 QAs)⁴⁹, and *MedXpertQA* (286 QAs)⁵⁰ - from 53 physician trainees. Physician trainees annotations are verified by a group of seven physician experts to ensure the reliability of their labels. We generate LLM relevancy labels in two ways. First by prompting all LLMs (GPT-4o⁵¹ and GPT-5⁵², MedGemma-27B⁵³, Qwen2.5-14B-Instruct, Qwen2.5-72B-Instruct⁵⁴, and Llama-3.1 70B-Instruct⁵⁵) to self-report sentence-level input relevance^{56,57}. We also use ContextCite⁵⁸ to retrieve sentence-level relevance from the four open-source LLMs tested.

Using these annotations, we compare LLMs and physician trainees in three ways. First, we measure the concordance rate on sentence level relevance. Board certified physicians agree with the physician trainee majority vote at a rate of 72.4%, while LLM concordance with the same reference ranges from 44.9% to 65.9%. Second, we examine how restricting the input to the sentences each labeller marked as highly relevant affects downstream QA performance, and find that most LLMs gain accuracy after this filtering, whether the filter is derived from physician trainee labels or from another LLM. Third, we quantify a *Spurious Rate* per model, defined as the fraction of originally correct answers that become incorrect once physician trainee identified low or irrelevant sentences are removed. This rate ranges from 5.8% to 24.8% in our sample and is best read as a proxy for reliance on content that physician trainees did not consider decisive, rather than as a direct measure of spurious reasoning. To our knowledge, *MedPAIR* is the first openly released benchmark that pairs sentence level physician trainee relevance labels with multiple LLM derived relevance labels on patient case QA, which is a complementary resource to rationale benchmarks such as ERASER that cover a broader range of tasks.

2 Results

2.1 Creating the MedPAIR dataset and evaluation framework

2.1.1 Medical QA Dataset Construction

To examine the alignment between physician trainees' and LLMs' assessments of relevance, we concentrate on existing QA pairs grounded in specific patient case scenarios, rather than general evidence-based questions. We draw on our four datasets *Massive Multitask Language Understanding* (MMLU)-precision medicine (released in September, 2020)⁴⁸, *Medbullets* (released in March, 2021, paper about Medbullets was preprinted in April 2025), *JAMA Clinical Challenge dataset* (collected during July 2013 to October 2023, paper about JAMA was preprinted in April 2025, required paywall access)⁴⁹, and *MedXpertQA* (released in January, 2025)⁵⁰. The characteristics for each dataset are presented in Supplementary Information. Each source provides multi-sentence patient case descriptions paired with questions (4-option or 10-option multiple-choice) and answers, offering a rich context for relevance annotation. Such patient vignettes are broadly recognized in the medical and social sciences, including health economics, and physician responses to clinical vignettes have been shown to predict realized billing behavior in the U.S. Medicare system⁵⁹. We obtained approval from the MIT Committee on the Use of Humans as Experimental Subjects (IRB: 2504001610) and preregistered our dataset collection protocol on the Open Science Framework (OSF)⁶⁰. No personally identifiable patient information was involved in this study.

Across the four datasets, there are 4,052 QA pairs in total. We first remove 2,052 pairs in which the patient profile contains two or fewer sentences. We then exclude 629 pairs that have one or no correct answers, as well as 448 pairs that require information from modalities other than text. This leaves us with a set of 933 QA pairs to obtain labels on. We use an online

Dataset	Data Source	Total QA	Total Options	Avg Sentence	Physician-Labeled Low/Irrelevant Sentence (%)
MMLU (Precision Medicine)	Medical textbooks	192	4	15.9 (7.0)	39.5%
JAMA Clinical Challenge	Journal articles	295	4	26.8 (8.5)	35.0%
MedBullets	USMLE	160	4	21.0 (4.6)	40.5%
MedXpertQA	Licensing Exam	286	10	14.9 (5.6)	32.8%
Overall		933	4/10	21.3 (8.8)	36.1%

Table 1. Comparative Analysis of Physician Trainee–Annotated *MedPAIR* Dataset Characteristics. Values in parentheses represent standard deviations.

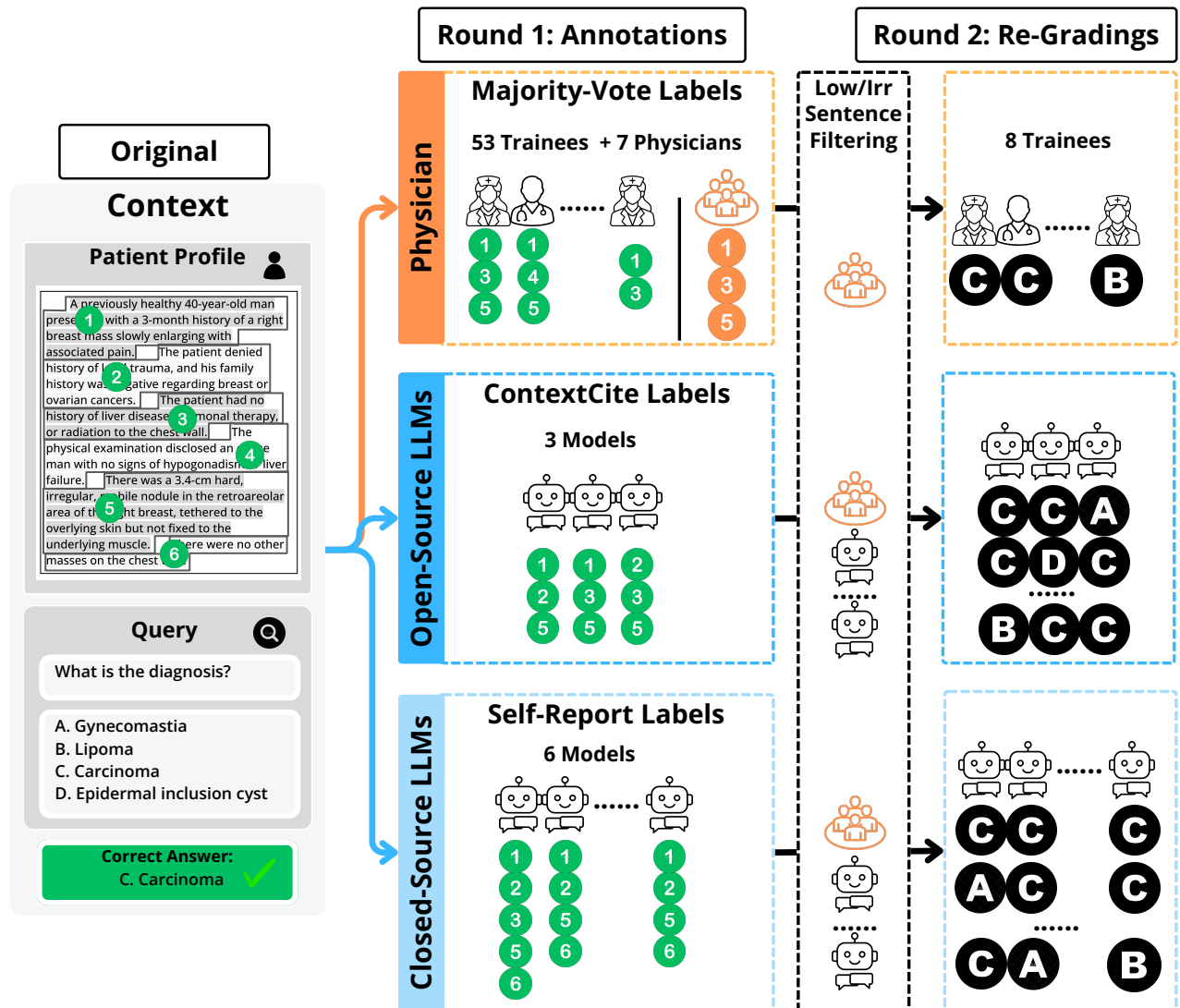


Figure 1. Study Design.

commercial labeling platform to recruit 52 physician trainees, generating a total of 6,224 QA labels on the 2,000 QA pairs. We remove all questions that no PT can answer correctly (448 pairs), leaving us with 933 QA pairs.

The four QAs have different characteristics, as JAMA Clinical Challenge are usually long and have lots of details, with

each sentence containing more words on average. The low relevance and irrelevant sentences also show the characteristics in perplexity that they are harder to predict as they are more complex, less structured, or diverge from typical language patterns the model has seen during training.

2.1.2 Physician Review of Physician Trainee Labels

Because board certified physicians could not annotate the full dataset, we used a smaller physician review to check whether the physician trainee labels were a usable reference for downstream analyses, rather than to construct an independent gold standard. To compare the two groups, we aggregated the trainee labels on each sentence into a single majority vote and compared it with the label assigned by a board certified physician. The average sentence level matching rate between the physician trainee majority vote and the reviewing physicians was 72.0% (median 73.7%, SD 17.7%, 95% CI [0.710, 0.730]). A cumulative distribution of these matching rates is presented in the Appendix. We treat this value as a measure of trainee physician agreement rather than of trainee accuracy, since the physicians themselves represent a small cohort and inter physician disagreement is not separately estimated in this review.

We refer to QA items for which the physician trainee physician matching rate falls below 20% as ambiguous cases and retain them in the dataset so that this subgroup can be analysed separately. Where the downstream analyses in this paper treat the physician trainee majority vote as a reference, this choice is a pragmatic one driven by scale, and the resulting concordance and filtering results should be interpreted against the 72.0% trainee physician agreement ceiling.

A board-certified physician, HJ, reviewed the physician-annotated majority-vote outcomes. Analysis of high- and low-relevance labels reveals that text marked as highly relevant by the physician trainee contains more anatomical structures and comparative descriptions (e.g., progressive, increased), whereas low-relevance text includes more historical information (medication, allergy, travel, social (smoking, illicit drug), etc.) and negative findings (uncomplicated, noncontributory, etc.).

From the full dataset, 30 QA pairs were randomly selected and Dr. HJ compared original and edited versions after removing irrelevant sentences, then categorized these removed low relevance or irrelevant sentences into thematic groups such as 1) *Redundant Clinical Details*, 2) *Negative Result* that is not essential for current chief complaint, 3) *Low relevant or Irrelevant Temporal Information*, 4) *History (Medical, Surgical, Medication, Social) with No/Very Low Clinical Information*, etc. The validation exercise evaluated whether the remaining highly relevant sentences maintained the link to the correct answer and whether removing low-relevance content affected answer correctness. The sample case study and validation sheet are presented in Supplementary Information.

We compared the final 933 QAs in *MedPAIR* on their average number of sentences, words per sentence, and perplexity. Across the four medical QA datasets, the highly relevant sentences are consistently longer and more uniform in structure, with lower perplexity values and thus greater linguistic predictability. In contrast, irrelevant or low-relevance sentences were shorter on average, displayed much higher variability in length, and proved more difficult for the language model to anticipate. The specific perplexity details are presented in the Supplementary Material.

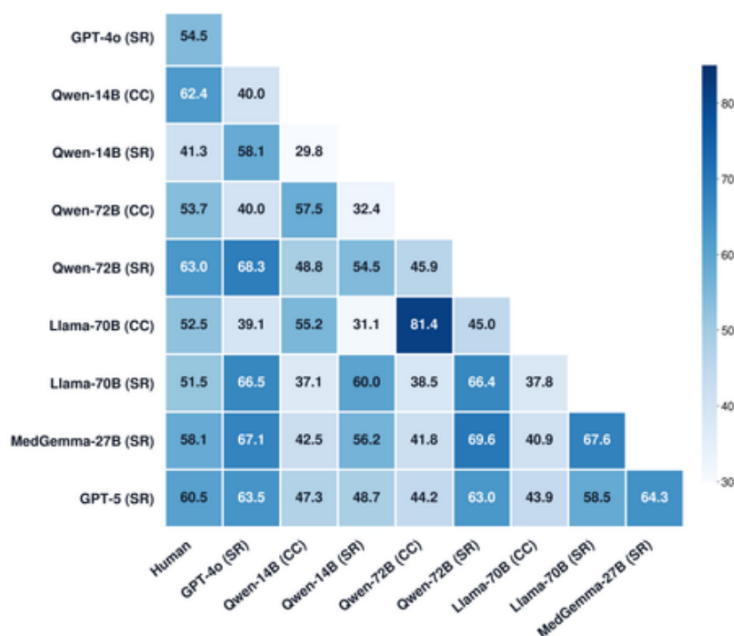
To compare the continuous ContextCite scores produced by the open source LLMs with the ternary labels assigned by the physician trainees, we mapped the numerical scores into the same three category scheme. For each QA pair, we let k equal the number of sentences marked highly relevant by the physician trainee majority vote. We then ranked the sentences by their ContextCite scores, assigned the top k to *high relevance*, and split the remaining sentences between *low relevance* and *irrelevant* using the corresponding trainee counts. This procedure aligns the LLM and human label distributions per item, which makes concordance directly interpretable in terms of which sentences each source considers most important. A consequence is that the total number of sentences placed in each category is held fixed across labellers for a given QA, so the resulting concordance rates reflect the ordering produced by each method rather than differences in how many sentences each method would otherwise label as relevant. We therefore treat concordance in this paper as a measure of ranking agreement, and discuss the sensitivity of concordance to alternative thresholding choices in the Supplementary Information.

2.2 Humans and LLMs Disagree on Information Relevance

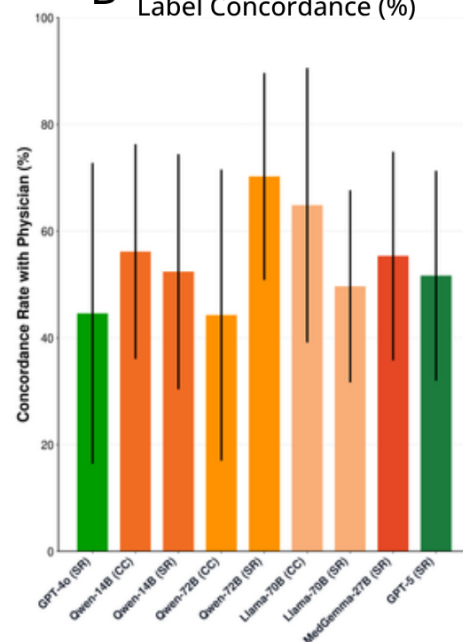
With the physician trainee labels characterised against a small physician review cohort and the LLM attribution scores mapped onto the same ternary scheme, we next asked how often physician trainees and LLMs agreed on which sentences carry weight for a given QA. This is an alignment question rather than a test of which labels are clinically correct, since the reviewing physicians themselves agree with the physician trainee majority vote only 72.0% of the time.

By examining cases in which physician trainees and LLMs produced differing relevance annotations, *MedPAIR* surfaces differences in how each source prioritises patient case information. We quantified the agreement between sentences marked highly relevant by the physician trainee majority vote and those highlighted by the models, using ContextCite scores for Qwen-14B, Llama-70B and Qwen-72B alongside GPT-4o self-reporting. The best performing model, Llama-70B, reached a mean agreement of 65.9%, below the 72.0% trainee physician agreement rate but not dramatically so. Roughly a third of sentences that physician trainees considered highly relevant were placed outside the top k sentences by at least one of the models. These gaps narrow the claim from one of stark misalignment to one of consistent but moderate disagreement, and

A Heatmap for LLMs Vs LLMs on Concordance Rate (%)



B Average Sentence-Level Label Concordance (%)



C Sample QA (MedXpertQA)

#	LLM 1	LLM 2	Physician	Patient Profile	Which of the following components is essential for culturing the most likely causative organism?
1	High	Low	High	1. A 20-year-old woman comes to the primary care clinic with complaints of a progressively worsening sore throat, dry cough, and ear pain.	A. Iron
2	Low	Low	High	2. She reports feeling mildly unwell for two weeks prior, with symptoms including fever, fatigue, muscle aches, and headache.	B. Urea
3	High	Irrelevant	Low	3. Her medical history is unremarkable, and her only daily medication is a combined oral contraceptive pill.	C. Magnesium
4	Irrelevant	Low	High	4. She works as a waitress at a local diner and mentions exposure to sick patrons.	D. Cysteine
5	Low	Low	Low	5. On examination, her temperature is 101.5°F, blood pressure is 118/76 mmHg, pulse is 78/min, and respiratory rate is 14/min.	E. Glucose
6	High	High	High	6. Chest auscultation is unremarkable, but a chest X-ray reveals bilateral multifocal opacities in the lungs.	F. Nicotinamide
7	Irrelevant	High	High	7. Sputum culture and Gram stain are negative.	G. Cholesterol (Correct Answer)
					H. Hemin
					I. Sulfate
					J. Pyridoxine

Figure 2. Average Sentence-Level Label Concordance (%). Panel A shows the sentence-level concordance between physicians and LLMs is highly variable across datasets and models. Physician Concordance is calculated as inter-rater reliability between physician and majority vote PT across all available labels (2-8 clinical labels). For LLMs, “CC” denotes the ContextCite score; “SR” denotes Self-Reported labels. Panel B shows a sample MedXpertQA QA, which is labeled by physician and LLMs on each sentence.

any downstream effect on QA accuracy must be read in that light. The full per-dataset and per-model breakdown is shown in Figure 2.

A common pattern was overattention to superficial cues. For example, a model might latch onto a laboratory value that is extreme and assume it must be important, even if it is not relevant to the question at hand. Conversely, models sometimes missed subtle but crucial cues that humans tagged as relevant. These findings could be partially due to LLMs occasionally attribute incorrect answers on misinterpreted or irrelevant context, indicating flawed input context relevance estimates. ContextCite highlights cases where a model justifies its answer by citing a sentence it wrongly deems supportive, what researchers term contributive attribution.

2.3 Human Relevance Improves LLM Performance

Given the moderate disagreement between physician trainees and LLMs on what counts as relevant, we next examined whether restricting models to physician trainee identified relevant sentences shifts their answers. Removing the sentences that physician

trainees marked as low relevance or irrelevant raised accuracy for most LLMs on most datasets. We read this as a filtering effect that is consistent with, but not uniquely explained by, reduced distraction from content the physician trainees did not consider decisive. Input length, exposure of memorised strings, and the deliberate inclusion of distractors in USMLE style items are also modified by this intervention, and we return to these confounds in the Discussion. Physician trainee labelling instructions explicitly stated that the QA should be answerable from the highly relevant sentences alone, so the filtering operation at least preserves the sentences that a physician trainee would identify as sufficient.

Figure 3 shows that removing sentences that physician trainees considered low relevance or irrelevant is associated with higher accuracy for most LLMs. The per dataset improvements vary in size and direction, and we report them here as observed differences rather than statistically tested effects, since the Round 2 subset contains only 248 QA pairs and we do not attach confidence intervals or significance tests to the individual per dataset changes in Figure 3. Qwen-72B's accuracy rises from 89.3% to 92.2% on MMLU Precision Medicine and from 35.1% to 62.2% overall, while GPT-4o moves from 95.6% to 96.4% and from 39.3% to 73.0%, keeping its position as the highest performing model before and after filtering. Similar directional changes appear on JAMA Clinical Challenge, MedBullets and MedXpertQA, with standard deviations below 0.5 in most cells. Qwen-14B and Llama-70B show small declines on the MMLU subset (marked in red), which is consistent with sensitivity to small input perturbations in weaker models. Large relative gains for MedGemma-27B are driven by low baseline accuracies and should be interpreted with caution: a change from a low baseline to a still moderate accuracy produces a large percentage change that is not necessarily clinically meaningful. Comparisons with physician trainee unfiltered accuracy (48.3%) are descriptive; the trainees and the models receive different inputs in the filtered condition and direct head to head claims about clinical parity are not supported by this design.

One interpretation of these results is that focusing on sentences that physician trainees regard as highly relevant reduces the influence of content they consider peripheral. Alternative interpretations are equally compatible with the data: shorter inputs are easier to process, filtering can accidentally remove adversarial distractors that USMLE style items include by design, and for datasets released several years before our evaluation (such as MMLU Precision Medicine, released in 2020) the filtered input may disrupt patterns associated with memorisation. Disentangling these mechanisms is beyond the scope of this study, and we discuss this limitation in the Discussion. Figure 4 shows that between 1.6% and 18.5% of originally correct predictions flip to incorrect after removing the sentences that physician trainees labelled low relevance or irrelevant. Among the models evaluated, Qwen-14B shows the highest *Spurious Rate* and closed source GPT-4o the lowest, with the caveat that this metric is not compared against a random sentence removal baseline and therefore also reflects general sensitivity to input perturbation.

2.4 LLM Relevance Improves LLM Performance

Physician trainee annotations are not practical to obtain for every deployment, so we next asked whether LLMs can select their own relevant sentences in a way that produces similar directional changes in performance. To address this, we evaluated the difference in task accuracy when each model was restricted to the sentences that it itself had labelled highly relevant, using per-sentence ContextCite scores from MedGemma-27B, Llama-70B, Qwen-14B, Qwen-72B, GPT-4o, and GPT-5.

Frontier models show limited sensitivity to this self filtering. GPT-5 exhibits almost no change across benchmarks, while GPT-4o and the Qwen models show small accuracy increases. Weaker and domain specialised models show larger relative changes, though, as noted above, large relative changes from low baselines should be read with caution. Similar directional patterns appear when one model filters another model's input. We do not claim that self filtering produces a new, higher performance regime; the more conservative reading is that input restriction based on a model's own attribution scores is not systematically harmful and can yield small gains on several benchmarks.

At the dataset level, JAMA Clinical Challenge remains the most difficult benchmark for all models, with accuracy between 3% and 33%, and relevance based filtering produces only modest changes there. These observations suggest that part of the gap between weaker and stronger models on *MedPAIR* is associated with how much each model is distracted by content that is not decisive for the question, although the effect is not large enough to overturn the existing model ordering.

3 Discussion

Our findings point to a consistent but moderate gap between the relevance judgments of physician trainees and those of current LLMs on clinical QA vignettes, with LLM concordance of 44.9% to 65.9% against a trainee-physician ceiling of 72.4%. This resonates with earlier work showing that LLM performance can be overestimated when accuracy is read in isolation⁶¹⁻⁶³. We interpret our result as evidence that accuracy metrics alone do not fully describe how LLMs arrive at their answers in this setting, rather than as a sharp dichotomy between aligned humans and misaligned models.

Better alignment between LLM selected and clinician selected evidence is a plausible ingredient of clinically deployable AI⁶⁴, but our study does not directly measure clinical deployment outcomes. Our items are multiple choice exam questions, and the sentences labelled as irrelevant in USMLE style vignettes often serve a pedagogical purpose. Real clinical notes contain redundant, noisy and sometimes contradictory content drawn from different distributions. Readers should therefore treat our

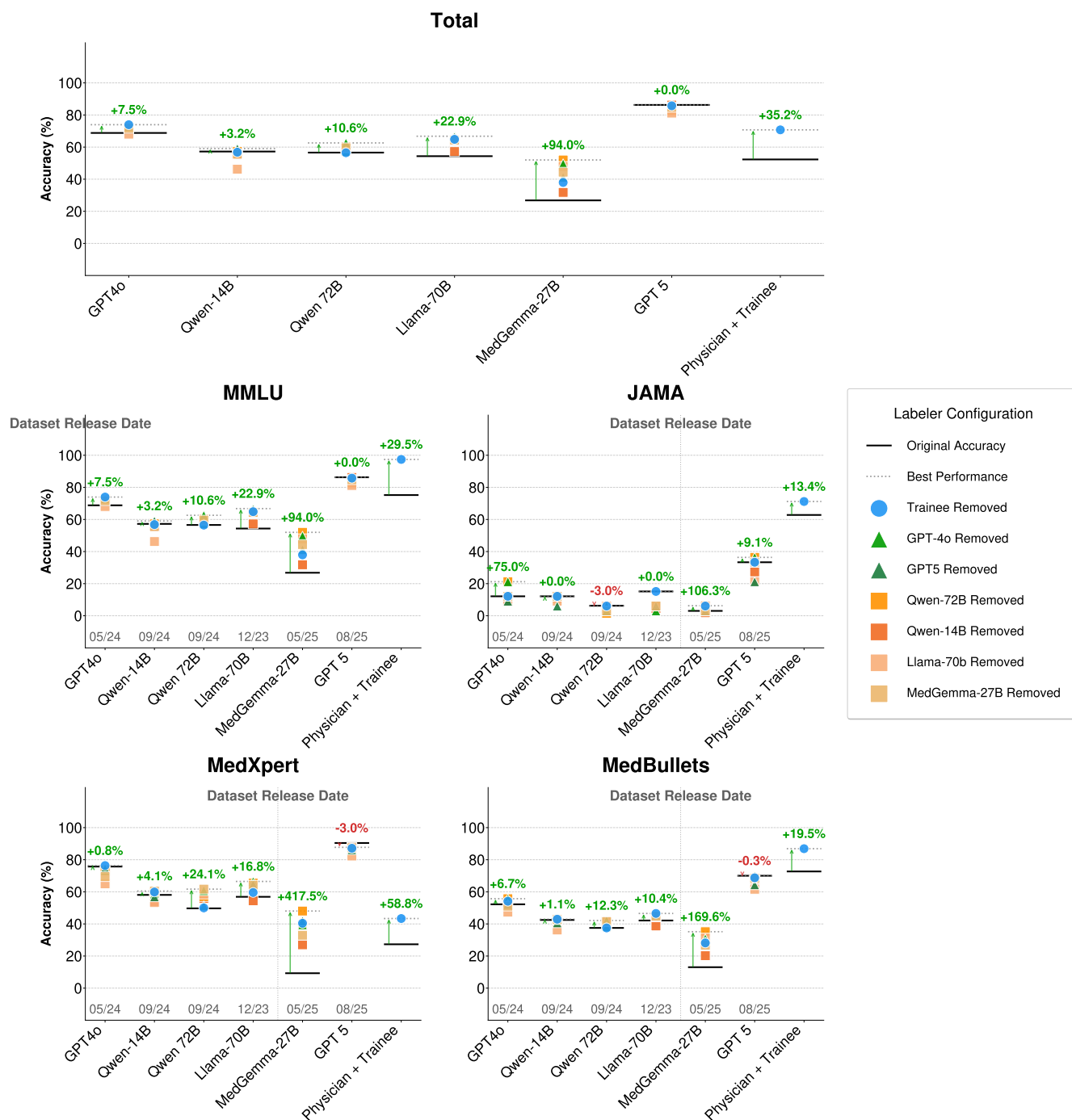


Figure 3. Effect of Filtering Context on Final Performance. Figure illustrates how accuracy shifts on four medical benchmarks (MMLU, JAMA, MedXpert, MedBullets) when utilizing various sets of LLMs-labeled highly relevant sentences. Solid black horizontal lines denote the original baseline accuracy, while dotted lines represent the highest accuracy achieved. Colored shapes indicate performance when specific models (GPT-4o, GPT-5, Qwen-14B and Qwen-72B, Llama-70B, MedGemma-27B, Trainee, Physician) highly-relevant sentences are used in predictions. Percentage annotations highlight the largest improvement or decline between the baseline and the best observed accuracy.

findings as an evaluation step that informs benchmark interpretation, and not as direct evidence about behaviour in live clinical workflows⁶⁵.

Previous work has demonstrated that selectively pruning input contexts and retaining only the most relevant context

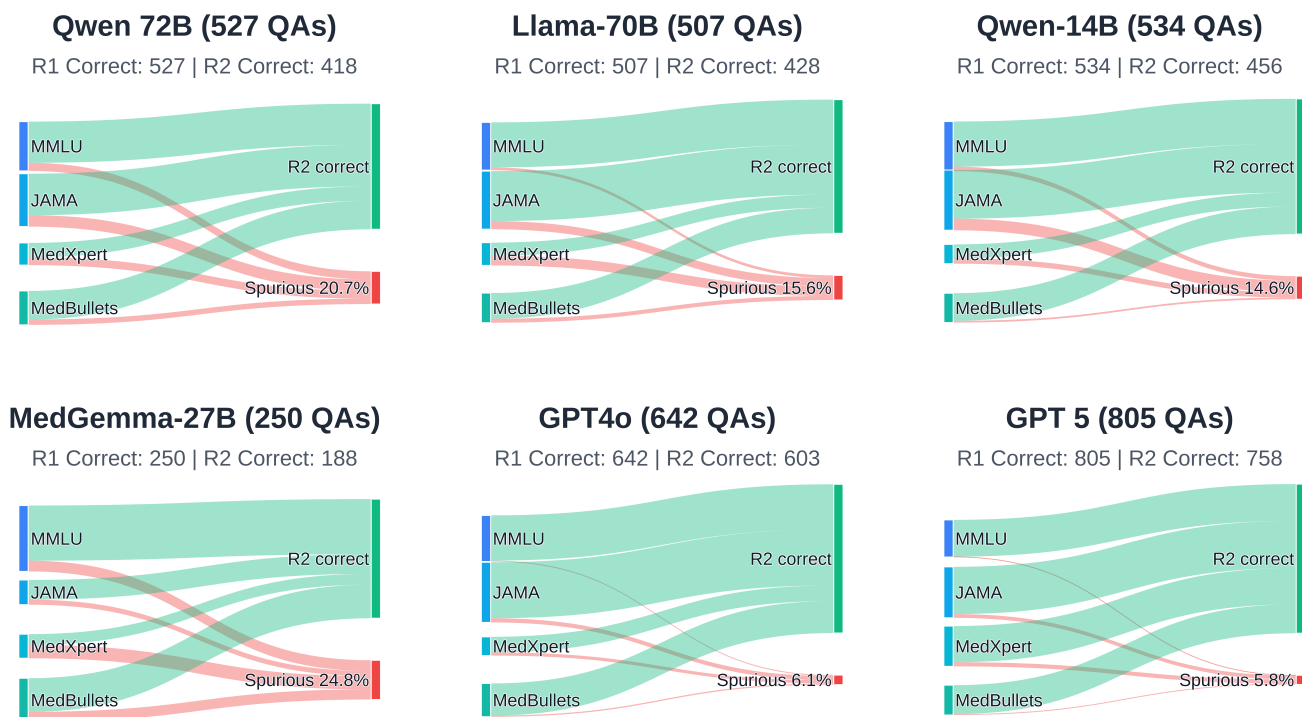


Figure 4. The Spurious Rate (%) of removing physician trainee-identified low-relevance and irrelevant sentences out of 933 QAs. Green bar indicates the proportion of QAs which remain correct in the second round, and the red bar indicates the proportion of QAs which become incorrect in the second round.

can enhance QA performance in language models^{66,67}. Our experiments extend these findings by showing that physician annotated context reduction, ContextCite scores from open-source models, or self-report prompting of close-source models each provides consistent performance gains across four medical QAs. *MedPAIR* addresses this need by introducing a fine-grained, expert-annotated benchmark for comparing model and physician relevance assessments. By establishing a physician-LLM relevance pair at the sentence level for each clinical case, *MedPAIR* enables direct comparison of reasoning processes and offers a foundation for evaluating how LLMs interpret and prioritize clinical information, an essential step toward developing models suitable for clinical deployment.

Additionally, the *MedPAIR* dataset contributes to understand whether LLM is able to automate the evaluation process as a judge. While our findings suggest that LLMs can enhance performance in this role, the substantial improvements observed with domain expert-generated datasets demonstrate the importance of human-in-the-loop during evaluation process^{68,69}. The human and LLM misalignment on information relevance highlight the need for expert oversight in ensuring accuracy^{70,71}. Although LLMs can provide ContextCite scores and self-report labels to explain the identification of input context, the quality and consistency of these outputs still require validation from human experts. This is particularly important in healthcare⁷², where automating prediction and evaluation with LLMs could have serious consequences due to potential misalignments with human judgment in input information retrieval⁷³.

This study has several limitations that future work should address. The human reference in our downstream analyses is a majority vote of physician trainees rather than an independent attending physician gold standard, and the 72.0% trainee-physician matching rate establishes an agreement ceiling that should be kept in mind when interpreting LLM concordance numbers. We did not separately estimate inter-trainee reliability with standard agreement statistics such as Fleiss' kappa or Krippendorff's alpha, and we treat the trainee majority vote as a pragmatic reference rather than a certified ground truth. The ternary High / Low / Irrelevant scheme is coarse by design, and our filtering experiments collapse Low Relevance and Irrelevant into a single removed category even though Low Relevance sentences can still support ruling out alternatives, which is a real diagnostic function. The ternary mapping from continuous ContextCite scores holds the per-item number of high relevance sentences fixed to match the trainee majority vote, so concordance measures ranking agreement conditional on that count rather than an absolute threshold; we discuss alternative mappings in the Supplementary Information.

Our filtering experiments do not disentangle reduced distraction from several confounded mechanisms, including shorter

input, disruption of patterns associated with training set memorisation on older benchmarks such as MMLU Precision Medicine (2020), and incidental removal of adversarial distractors that USMLE style items include by design. We did not include a matched random sentence removal baseline, and without this baseline the *Spurious Rate* cannot be uniquely attributed to reliance on content that clinicians would consider peripheral^{74–77}. The Round 2 filtering analysis is based on 248 QA pairs and we report observed accuracy differences rather than confidence intervals or significance tests; large relative gains for models with very low baselines, such as MedGemma-27B, should be read in absolute rather than relative terms. Our model panel mixes open and closed source systems at different scales and is not exhaustive, and the omission of Med-PaLM, Claude and other medically fine tuned models limits the breadth of cross model comparisons. Finally, *MedPAIR* is built from multiple choice exam QA, and the relevance patterns in exam vignettes may differ systematically from those in real clinical notes^{78,79}. We release *MedPAIR* as an evaluation resource for studying these questions and encourage future work that re-runs our filtering analyses with matched random and clinician-varied baselines, contamination audits, finer grained label schemes, and broader model coverage.

MedPAIR provides a pre-reasoning evaluation resource that quantifies sentence level alignment between LLM relevance judgments and physician trainee annotations across four medical QA benchmarks. We describe relevance pairs as a way to ask which parts of a case each source treats as central, and we use these pairs to flag systematic mismatches in how LLMs process patient context. Across 933 annotated QA items, restricting each LLM to the sentences that physician trainees marked as highly relevant produced consistent directional accuracy gains, while the residual *Spurious Rate* identifies cases in which originally correct answers are sensitive to this restriction. Taken together with the limitations above, these results position *MedPAIR* as a diagnostic benchmark for studying how LLMs attend to clinical inputs, rather than as a direct measure of readiness for clinical deployment.

4 Methods

4.1 Expert Data Annotation

We partner with Centaur Labs (<https://centaur.ai>) to employ board-certified physicians and physician trainees (medical students or higher qualifications) annotate the QA pairs. Physician trainees are chosen for their familiarity with medical exam preparation, as these questions are primarily designed for medical students. The demographic information is presented in the Supplementary Information.

A total of 52 physician trainees participated (mean age: 26.9), with 77.3% at the advanced training level and 67.4% of the labelers had passed the United States Medical Licensing Examination (USMLE) Step 1 Exam. Notably, half of them reported familiarity with using LLMs in clinical contexts, such as integrating tools like ChatGPT into clinical queries or workflows. On average, each labeler spent an average of 3.15 minutes (SD 2.90, 95% CI: 3.04–3.25, n = 2952) per QA. Medical students spent on average 3.19 minutes (SD 3.16, 95% CI: 3.21–3.44, n = 2018); physicians spent on average 3.05 minutes (SD 2.25, 95% CI: 2.90–3.19, n = 934). For each case, physician trainees first selected the most appropriate answer and then annotated the relevance of every sentence in the patient profile.

A total of 7 physicians participated (mean age: 43.4, 4 males, and 3 females), with an average of 17.0 years of post-graduate year. All physicians passed USMLE step 3 or equivalent exams, with 2 physicians from United Kingdom. The physician cohort showed high familiarity with LLMs in healthcare, with 5 out of 7 physicians reported positive that LLM deployment is ready for clinician-in-the-loop tasks.

A sample physician trainees' annotation is presented in Figure 1 (orange boxes). Full study instructions and the pre- and post-survey instruments are provided in the supplementary material. Data collection occurred between March 6 and May 5, 2025.

Each QA is annotated by at least three physician trainees and verified to contain at least one correct answer. For each sentence, annotators applied one of three labels: (1) **High Relevance:** Information that is critical and must be considered to answer the question correctly. These are the key clinical clues or data points that strongly point toward the correct diagnosis or decision. (2) **Low Relevance:** Information that provides some context or minor clues but is not essential. These details might help rule out alternatives, yet the question could still be answered correctly without them. (3) **Irrelevant:** Information that is not pertinent to determining the correct answer. These can be distractors or background details included in the vignette that do not impact the outcome in the given context.

This annotation process presents a fine-grained ground truth of relevance for every QA: a trinary label for each sentence in the context, representing the physician trainee consensus on whether that piece of information is pertinent to the question. While obtaining these annotations demanded expert effort, they serve as a gold standard for capturing what physicians deem significant. This level of detailed expert labeling is largely absent from existing medical QA benchmarks, which typically include only the question and answer, without explicit identification of supporting case details and their degree of relevance⁴⁹.

4.2 LLM Data Annotations

We access the closed-source models (OpenAI) through their respective Python API libraries between March to October 2025. For the open-source models (Llama, Qwen, and Meditron), we employ the Hugging Face Transformers library. These models are executed on a local cluster equipped with four NVIDIA A100 GPUs (80 GB each). All models run with default hyperparameter.

To directly compare the majority-vote annotations of the physician trainees with the labels generated by LLM for each pair of QA, we annotated the LLM outputs using both ContextCite with open-source models and a self-report prompt with close-source models. ContextCite scores approximate the model’s attention distribution across sentences⁸⁰, while self-reported labels capture the model’s own assessment of sentence relevance via prompting.

Then we performed a sentence-level analysis of their respective annotations to examine this divergence at the sentence level. We examined one closed-source model GPT-4o⁸¹ and three open-source models: Qwen-14B, Llama 3.1 Instruct 70B⁵⁵, and Qwen 2.5-72B⁵⁴. For GPT-4o, we structured the study the same as the physician trainee labeling protocol: we fed the identical instruction prompt three times and determined the relevance label for each sentence by majority vote. For the open-source models, we applied ContextCite to quantify the relevance of each sentence within the QA contexts, as ContextCite provides a simple, scalable mechanism for tracing portions of a generated response back to specific input sentences⁵⁸. Each model received the identical prompt used by the physician labelers to elicit sentence-level relevance judgments. The complete prompts for generating self-reported labels and ContextCite annotations are provided in Supplementary Information.

4.3 Problem Formulation

Suppose that a particular question consists of a set of sentences $\mathcal{S} = \mathcal{S}^+ \cup \mathcal{S}^-$, where $\mathcal{S}^+$ is the set of relevant sentences (as labeled by a physician trainee), and $\mathcal{S}^-$ is the set of irrelevant sentences. Let $Y \in \{1, 2, \dots, K\}$ be the true label. Suppose we have some LLM $f : 2^{\mathcal{S}} \rightarrow \{1, 2, \dots, K\}$.

In order to probe whether f answers the question using the same information as a human, we compare $f(\mathcal{S})$ with $f(\mathcal{S}^+)$, under the assumption that the set $f(\mathcal{S}^+)$ is sufficient for a human to answer the question correctly. This gives us the following possible scenarios:

1. $f(\mathcal{S}) = f(\mathcal{S}^+) = Y$: The model is correct in both cases, suggesting that it relies on the same information as a human to solve the problem.
2. $f(\mathcal{S}) \neq Y, f(\mathcal{S}^+) \neq Y$: The model is incorrect in both cases, indicating that the problem is inherently difficult or f has poor capabilities.
3. $f(\mathcal{S}) = Y, f(\mathcal{S}^+) \neq Y$: By removing irrelevant sentences, we flip a correct prediction to an incorrect one. This indicates that the model may have been relying on spurious information in $\mathcal{S}^-$ (i.e. information for which a human deems irrelevant) to make its predictions.
4. $f(\mathcal{S}) \neq Y, f(\mathcal{S}^+) = Y$: The model improves when irrelevant information is removed, indicating that $\mathcal{S}^+$ contains sufficient information to answer the question as expected, and the presence of $\mathcal{S}^-$ introduces noise or distractions.

As case (3) is the most salient for our purpose, we define the **Spurious Rate** (**Spurious Rate**) as the prevalence of samples in this case:

$$\text{Spurious Rate}(f) = \frac{\sum_{i=1}^N \mathbf{1}[f(\mathcal{S}_i) = Y_i \wedge f(\mathcal{S}_i^+) \neq Y_i]}{\sum_{i=1}^N \mathbf{1}[f(\mathcal{S}_i) = Y_i]},$$

where N is the total number of questions. We treat **Spurious Rate** as a proxy rather than a direct measure of spurious reasoning. Removing sentences changes the input distribution, and any model may behave differently on a truncated input regardless of whether the removed content was truly non-informative. The physician trainee labels that define $\mathcal{S}^-$ are themselves noisy, so the metric also reflects sensitivity to label noise. A fully controlled interpretation would require a random sentence removal baseline matched for length, which we leave to future work. With these caveats in mind, a higher **Spurious Rate** indicates that more of a model’s originally correct answers do not survive the removal of sentences that physician trainees judged low relevance or irrelevant, while a lower value indicates more robustness to this particular perturbation.

5 Data Availability

All data underlying this study’s results are provided in the Supplementary Data. All LLM and physician trainee-labeled data are available at: <http://medpair.csail.mit.edu/>. The physician trainee-annotated data are available to use through HuggingFace at <https://huggingface.co/datasets/YuexingHao/MedPAIR>.

6 Code Availability

All codes generating this study's results are presented in the Supplementary Data.

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8 Author Contributions

Y.H., K.A., H.Z., and M.G. conceived and designed the study. Y.H., K.A., G.Y., and H.J. carried out data collection, processing, and experimental analysis. H.Z., G.Y., I.P., P.T., M.S., S.K., and A.D.S. offered critical intellectual input and guidance. Y.H., H.Z., H.J., and M.G. drafted the initial version of the manuscript. M.G. secured funding and provided overall supervision. All authors contributed to administrative, technical, and material support, reflecting a comprehensive and collaborative team effort.

9 Competing Interests

The authors declare that they have no competing interests.

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